

Accountable AI: Best Practice Guide

The following table outlines a sample of emerging best practices and examples from accountable AI programs used by organizations from different sectors, geographies, and sizes. These practices are mapped to the corresponding element of the CIPL Accountability Framework. The practices are not intended to be mandatory industry standards but rather serve as examples of how companies are implementing specific practices to foster accountability in their development, deployment, and use of AI technologies. Each of the following should be calibrated based on risks, industry context, business model, size, and maturity level of the organization.

Leadership and Oversight

ACCOUNTABILITY ELEMENT	RELATED PRACTICES
Leadership and Oversight	- Establishing "tone from the top" and demonstrating a commitment to advance ethics, values, and specific principles in AI development, deployment, and use - Implementing systematic processes and escalation pathways for AI-related decision making - Establishing AI ethics and oversight bodies or committees (internal or external) to review risky AI use cases and promote ongoing improvements to AI practices - Appointing a board member for AI oversight - Appointing a responsible AI lead, AI officer, or AI champion - Setting up an internal interdisciplinary AI board or AI committee - Establishing organization-wide AI ethics principles - Ensuring inclusion and diversity in AI model development and AI product teams - Creating a centralized governance framework with oversight from the top that still provides flexibility within internal teams - Expanding the remit of privacy teams to include AI-related responsibilities - Leveraging the expertise of other relevant teams (e.g., engineering, data science, legal, ethics and compliance, etc.) to ensure multidisciplinary, cross-functional AI teams - Encouraging employee reporting throughout all levels of the organization by offering escalation pathways to resolve potential AI-related issues

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Risk Assessment

ACCOUNTABILITY ELEMENT	RELATED PRACTICES
Risk Assessment	• Developing algorithmic impact assessments or fairness assessment tools to monitor and continuously test algorithms to avoid human bias, unfair discrimination, and "concept drift" throughout the entirety of the AI lifecycle • Requiring AI risk assessments at multiple points throughout the AI lifecycle, particularly for a new or updated use cases or applications • Creating ethics, human rights, and/or data protection impact assessments • Creating a risk taxonomy that categorizes AI-related risks and allows for uniform assessment • Keeping a centralized repository of all risk assessment documentation • Developing standardized risk assessment methodologies that consider the benefits of the AI applications or use, the likelihood and severity of risk factors on individuals and/or society, the level of human oversight needed for individually automated decisions with significant impact (e.g., legal ramifications), the ability to explain the technology in the appropriate context, and the ability to audit its effectiveness • Documenting considerations (e.g., accuracy, data minimization, security, transparency, scope of impact, benefits to society) for high-risk processing • Assessing data quality against key performance indicators (KPIs) • Evaluating the data vis-à-vis the purpose of its use (i.e., the quality of the data, its provenance, whether it's personal, synthetic, in-house, or externally sourced) • Developing frameworks for data preparation and model assessment - including feature engineering, cross-validation, back-testing, standardized KPIs • Enabling close collaboration between business and data experts (e.g., data analysts, data engineers, IT, and software engineers) on a regular basis to assess accuracy, ensure appropriate outputs and allow for proper use of the model • Using privacy enhancing technologies (PETs) to preserve the privacy and security of AI systems • Outlining escalation pathways to send AI-related issues to an AI ethics council or other oversight body

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Policies and Procedures

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Policies and Procedures	• Adopting specific AI policies and procedures on how to develop, deploy, or sell AI • Drafting policies on the application of privacy and security by design principles throughout the AI lifecycle • Setting rules on the level of verification for data input and output • Requiring pilot testing of models before release • Specifying the use of protected data (e.g., encrypted, pseudonymized, tokenized, or synthetic data) in training AI models • Creating a glossary of AI-related terms for internal use and reference • Promoting the use of smaller, higher quality data sets • Cleaning and curating data sets before model training through automated or manual checks • Considering relevant and appropriate use of PETs and PPTs to integrate privacy and security controls into AI models • Outlining special considerations for organizations creating and selling AI models, software, applications • Developing a fairness or AI impact assessment to analyze and mitigate AI-related risks • Creating due diligence/self-assessment checklists or tools for business partners deploying AI • Clearly defining escalation steps for reporting high-risk AI issues • Implementing an ideation phase with all stakeholders (e.g., data scientists, business, final user, control functions) where needs (including explainability), outcomes, validations rules maintenance, and budget are discussed • Implementing specific policies for internal GenAI use • Requiring consideration for diversity in relevant teams and business functions • Implementing internal policies in parallel with forthcoming AI regulation • Translating internal principles-based policies to third-party vendor agreements, language, and due diligence processes • Creating processes for review of high-risk AI use cases by an AI ethics board or council

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Transparency

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Transparency	• Tailoring transparency measures for the different needs of end users, regulators, business partners, and internal stakeholders at all stages of the AI lifecycle • Communicating disclosures in a simple, easy-to-understand manner • Considering how AI disclosures can be inclusive and accessible for those with special needs/disabilities • Establishing a transparency trail to explain automated decision-making and broad workings of algorithms • Providing notice when the system relies on AI/ML • Providing counterfactual information (e.g., how different inputs can affect the output of an AI model) • Understanding customers' expectations and deploying AI technologies based on their readiness to embrace AI • Implementing tiered transparency • Defining criteria for internal deployment of AI technologies based on usage scenarios and communicating them to users • Publishing model or system cards (i.e., short documents accompanying AI models that describe the context in which a given model is intended to be used and how the model performs in a variety of conditions) • Creating a data hub for information regarding data governance, data accessibility, data lineage, data modification, data quality, etc. • Tailoring transparency to the identified risk (e.g., using watermarking for generative AI output) where possible and appropriate • Participating in benchmarking opportunities, public engagement, and regulatory sandboxes • Using visualization tools to depict difficult, technically complex concepts to end users

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Transparency

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Training and Awareness	• Providing specific training for data scientists and engineers, including how to address relevant ethical issues (e.g., how to limit and address bias) • Creating opportunities for cross-functional training (e.g., between privacy professionals and AI engineers) • Tailoring trainings regarding ethics and fairness in AI for relevant teams • Compiling and making available AI use case information where relevant risks have been mitigated or deployment has been halted • Creating a "translator" role that helps explain the the impact and technical capacities and limitations of AI • Sharing case studies to help employees learn how to address potentially complex, ethically challenging AI cases • Incentivizing compliance with completing ethics training by pairing it with eligibility for bonuses, pay rises, and/or promotions, or incorporating it into other mandatory training activities

Response and Enforcement

ACCOUNTABILITY ELEMENT	RELATED PRACTICES
Response and Enforcement	• Enabling redress mechanisms to remedy an AI decision • Permitting redress through a human, not to a bot • Developing communication channels for internal (e.g., for employees) and external (e.g., end users, business customers) to report and address feedback, complaints, requests, etc.

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Monitoring and Verification

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Monitoring and Verification	• Incorporating "human in the loop" (HITL) in design, oversight, and redress • Identifying and understanding which business functions are using AI • Providing the capability for human audit of input and output • Ensuring human review of individual decisions with legal or similarly significant effects • Monitoring the data ecosystem - from data flow in, through data process, to data flow out • Using different auditing techniques • Deploying counterfactual testing techniques • Pre-defining AI audit controls • Creating an internal audit team with expertise in AI and other emerging technologies • Allowing human control or intervention where technically possible and reasonable necessary • Monitoring AI models (e.g., back-testing and feedback loop) and conducting ongoing maintenance • Red teaming and adversarial testing of AI models